

When Does Message Passing Help? A Small Controlled Study on Contextual Stochastic Block Models

Heron-30R

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/gcn-homophily-sbm>

Abstract—We ask when message passing beats a feature-only classifier, using contextual stochastic block model graphs (3 classes, 900 nodes, mean degree 10) where edge homophily h runs from 0 to 1 and feature informativeness $\mu \in \{0.5, 1, 2\}$. We compare four reimplemented, plain-torch systems: an MLP, a 2-layer GCN, a GCN with separate self and neighbour weights (the H2GCN idea) and label propagation, over 33 cells with 5 seeds each (660 runs). GCN falls below the MLP in a band of low-to-mid homophily, but the band sits around $h = 0.2$ – 0.3 rather than only at the middle; at $\mu = 1$ the deficit is significant at $h = 0.3$ (paired t -test, 5 seeds) and not at $h = 0.4$, so our pre-registered mid-homophily hypothesis is only partly supported. The dip deepens as features get stronger. Separating self and neighbour weights removes most of the dip and is by far the best system under strong heterophily, but it is not uniformly at least as good as GCN or MLP: it trails the better of the two by more than 0.02 in 6 of 33 cells, and a tied-weights ablation collapses accuracy at $h = 0.1$. Label propagation wins only at the highest homophily and we found it also wins at stronger features, contradicting our hypothesis. This is a small CPU study on one synthetic family with fixed degree, class count and untuned depth, so the maps should be read as controlled illustrations, not as claims about real graphs.

Index Terms—graph neural networks, homophily, heterophily, stochastic block model, label propagation, node classification

I. Introduction

Graph neural networks (GNNs) are usually motivated by homophily: neighbouring nodes tend to share labels, so averaging over neighbours denoises features. The graph convolutional network (GCN) of Kipf and Welling [1] is the canonical example, and a line of work on heterophily argues that it should fail when neighbours have different labels and proposes designs such as separating the ego embedding from the neighbour embedding [2] or learning signed propagation weights [3]. Yet the evidence is mixed. Ma et al. show that GCN can do well on some heterophilous graphs [4], Luan et al. argue that homophily is the wrong summary statistic and that post-aggregation node similarity matters more [5], and Platonov et al. find

that standard GNNs usually beat specialised heterophily models once benchmark flaws are fixed [6].

On real benchmarks, homophily is confounded with feature quality, degree, graph size and split protocol, which makes it hard to say which factor drives a result. A contextual stochastic block model (CSBM) [7] removes the confound: it generates a graph with a planted community structure together with Gaussian node covariates, so one can vary edge homophily and feature informativeness independently. Theory for this family shows that graph convolution widens the regime in which the classes are linearly separable under homophily [8]; it is less clear what happens in between, where the graph is nearly uninformative and averaging mixes classes.

This paper is a small, controlled empirical map of that question. We sweep homophily from 0 to 1 and feature strength over three levels, and compare an MLP, a 2-layer GCN, a GCN with separate self and neighbour weights, and label propagation, all written from scratch in torch. We register five hypotheses before running the main grid (Section IV) and report each verdict, including the ones that fail. Our contributions are: (i) a win-region map over 33 homophily $\times$ informativeness cells with 5 seeds each; (ii) a test of the claim that GCN drops below the MLP at mid homophily, which is supported only partially and localised below the middle; (iii) an ablation showing that the ego/neighbour separation, not extra capacity, is what matters under heterophily; and (iv) a mean-degree sensitivity check at one cell. The study is deliberately small (900 nodes, 2 layers, CPU, 5 seeds), and we state what it cannot show in Section VII.

II. Related Work

Message passing and homophily. GCN [1] aggregates symmetrically normalised neighbourhoods including a self loop, which acts as a low-pass filter. Oono and Suzuki show that very deep GCNs lose expressive power [9]; we keep two layers throughout, so depth effects are out of scope.

Heterophily-oriented designs. Zhu et al. identify ego/neighbour separation, higher-order neighbourhoods and combining intermediate representations as design

Heron-30R is the codename of an autonomous study, not a person. Produced by the Stemma autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

choices that help under heterophily [2]; we reimplement only the first (and optionally 2-hop neighbourhoods) in a simplified form, so our “H2GCN-style” model is not the published architecture. GPR-GNN learns signed propagation weights [3], and LINKX shows that models that treat features and structure separately are strong on large non-homophilous graphs [10]; we do not run either. The widely used Geom-GCN heterophily datasets [11] are, as Platonov et al. document, affected by duplicate nodes, which motivates our use of synthetic data [6].

When does a GNN lose to an MLP? Ma et al. argue that homophily is not necessary for GCN to work and tie performance to how distinguishable the neighbourhood label distributions are [4]; Luan et al. propose post-aggregation similarity metrics to explain cases where GNNs lose to MLPs [5]. Our experiment can be seen as a controlled version of this question in which the only things that change are the edges (with the features, labels and split of each seed held fixed across h).

CSBM theory. Deshpande et al. introduced the contextual SBM and characterised information-theoretic limits of combining graph and covariates [7]; Baranwal et al. analyse one graph convolution on a two-class version and show that it extends the linearly separable regime under homophily [8]. We use the generative model but only measure finite-sample accuracy of trained models; we make no theoretical claim.

III. Method

Data model. There are $n = 900$ nodes, $K = 3$ balanced classes y_i and $nd/2$ sampled edges ($d = 10$ unless stated). For each edge a source node is drawn uniformly; the target is drawn from the source’s own class with probability h and otherwise uniformly from one of the other $K - 1$ classes; self loops are dropped and duplicates collapse, so realised mean degree is slightly below d and realised edge homophily deviates slightly from h (it is logged for every run as `realized_h`). Features are $x_i = \mu m_{y_i} + \epsilon_i$ with m_c random unit vectors in $\mathbb{R}^{16}$ and $\epsilon_i \sim \mathcal{N}(0, I)$. The uninformative-graph point is $h = 1/K$. For a fixed seed the features, labels and split are identical for all h (only edges change).

Notation. A is the symmetric adjacency matrix, $\tilde{A} = A + I$, $S = D^{-1/2}AD^{-1/2}$ and $\tilde{A} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$.

MLP. $\hat{Y} = W_2 \sigma(W_1 X)$ with dropout on the inputs of both layers.

GCN (reimplemented). $H^{(1)} = \sigma(\hat{A}XW_1)$, $\hat{Y} = \hat{A}H^{(1)}W_2$ [1].

H2GCN-style (reimplemented). Each layer keeps a separate self weight and neighbour weight and no self loop in the aggregation:

$$H^{(l+1)} = \sigma(H^{(l)}W_{\text{self}}^{(l)} + S H^{(l)}W_{\text{nb}}^{(l)}), \quad (1)$$

two layers, the last without σ . This keeps only the ego/neighbour separation of [2]; it omits the intermediate-representation concatenation. Two ablation flags: tied

weights $W_{\text{nb}} = W_{\text{self}}$, and adding a strict 2-hop term $S_2 H W_{\text{nb},2}^{(l)}$ where S_2 is the symmetric normalisation of the pairs at distance exactly two.

Label propagation (reimplemented). With Y_0 the one-hot training labels (zero rows elsewhere), iterate $F \leftarrow \alpha S F + (1 - \alpha) Y_0$ for 50 steps and predict $\arg \max_c F_{ic}$, with $\alpha \in \{0.5, 0.9, 0.99\}$ selected on validation accuracy. It uses no features.

Training. Full-batch Adam on cross entropy over the training nodes, 300 epochs with early stopping (patience 100) on validation accuracy; the reported test accuracy is taken at the best-validation epoch.

IV. Experimental Setup

Tasks. A task is a cell (h, μ) with $h \in \{0, 0.1, \dots, 1\}$ and $\mu \in \{0.5, 1, 2\}$ (33 cells). For each seed in $\{0, \dots, 4\}$ we regenerate graph, features, split (random 20%/20%/60% train/validation/test) and initialisation. Systems: MLP, GCN, H2GCN-style, label propagation (LP), all reimplemented in plain torch (no PyG), giving 660 main runs, all with the same metric (test accuracy), data and training budget.

Tuning. Learning rate $\in \{0.01, 0.05\}$, dropout $\in \{0, 0.5\}$ and weight decay $\in \{5 \times 10^{-4}, 5 \times 10^{-3}\}$ (8 configurations) were scored by validation accuracy averaged over cells $h \in \{0.2, 0.5, 0.8\}$ at $\mu = 1$ on separate tuning graphs (seeds 100–102); each trained model gets the selected triple (MLP: 0.01/0.5/5×10^{−3}; GCN: 0.05/0.5/5×10^{−4}; H2GCN-style: 0.05/0.5/5×10^{−3}). Hidden size 32 for all models and LP’s α is picked per run on validation accuracy. The grid is small, and differences between configurations on validation were small (log in `experiments/tuning_log.json`); we did not tune depth or hidden size.

Hypotheses and tests (registered in `proposal.md` before the main runs). H1: at $\mu = 1$ and $h \in \{0.3, 0.4, 0.5\}$ GCN is below the MLP (paired two-sided t -test over 5 seeds per cell, $\alpha = 0.05$, uncorrected). H2: H2GCN-style is within 0.02 of or above $\max(\text{MLP}, \text{GCN})$ in the mean over seeds in every cell. H3: LP is best only for $h \geq 0.8$ with weak features ($\mu = 0.5$) and is below the MLP for $h \leq 0.5$. H4: the crossover h above which GCN exceeds the MLP is lower for $\mu = 0.5$ than for $\mu = 2$. H5: tying self and neighbour weights hurts H2GCN-style at $h \in \{0.1, 0.4\}$, $\mu = 1$ (paired t -test, 5 seeds). H4 as originally written was “the lowest h at which GCN exceeds the MLP”; because GCN already exceeds the MLP at $h = 0$ in this model, the literal reading is trivially false and we evaluate the crossover on the homophilous branch $h \geq 1/3$ (a post-hoc clarification, flagged again in Section V).

Additional studies. (a) An ablation of H2GCN-style with tied weights and with a 2-hop term, at $\mu = 1$ and $h \in \{0.1, 0.4, 0.9\}$ (15 runs per variant; the full model’s rows come from the main grid). (b) A mean-degree sweep $d \in \{2, 5, 20\}$ for MLP, GCN and H2GCN-style at $h = 0.4$, $\mu = 1$ (45 runs; $d = 10$ rows from the main grid).

TABLE I

Test accuracy at $\mu = 1$ (mean $\pm$ std over 5 seeds; bold = best mean); last column: paired p for GCN vs MLP.

h	MLP	GCN	H2GCN	LP	$p_{\text{GCN-MLP}}$
0.0	0.539 $\pm$ 0.029	0.709 $\pm$ 0.039	0.911 $\pm$ 0.025	0.321 $\pm$ 0.022	4.070×10^{-5}
0.1	0.539 $\pm$ 0.029	0.542 $\pm$ 0.034	0.724 $\pm$ 0.037	0.327 $\pm$ 0.008	0.6512
0.2	0.539 $\pm$ 0.029	0.444 $\pm$ 0.020	0.587 $\pm$ 0.023	0.311 $\pm$ 0.038	0.0007776
0.3	0.539 $\pm$ 0.029	0.449 $\pm$ 0.023	0.535 $\pm$ 0.039	0.312 $\pm$ 0.027	0.007293
0.4	0.539 $\pm$ 0.029	0.506 $\pm$ 0.023	0.536 $\pm$ 0.040	0.389 $\pm$ 0.023	0.1424
0.5	0.539 $\pm$ 0.029	0.638 $\pm$ 0.031	0.635 $\pm$ 0.034	0.501 $\pm$ 0.022	0.01355
0.6	0.539 $\pm$ 0.029	0.792 $\pm$ 0.024	0.770 $\pm$ 0.026	0.664 $\pm$ 0.016	0.0001414
0.7	0.539 $\pm$ 0.029	0.904 $\pm$ 0.011	0.890 $\pm$ 0.018	0.838 $\pm$ 0.023	1.535×10^{-5}
0.8	0.539 $\pm$ 0.029	0.970 $\pm$ 0.010	0.967 $\pm$ 0.010	0.960 $\pm$ 0.014	2.537×10^{-6}
0.9	0.539 $\pm$ 0.029	0.992 $\pm$ 0.007	0.994 $\pm$ 0.003	0.996 $\pm$ 0.003	3.947×10^{-6}
1.0	0.539 $\pm$ 0.029	0.997 $\pm$ 0.003	0.998 $\pm$ 0.002	1.000 $\pm$ 0.000	4.244×10^{-6}

Compute. A shared CPU machine, 2 threads, no GPU; one training run takes about one second (LP much less), and the whole main grid took on the order of tens of minutes of wall clock including process start-up. An accidental double launch of the main grid produced duplicate registry entries and one failed entry (a file-name collision between the two launches); runs are deterministic, the duplicates were verified identical in accuracy and were removed, leaving exactly 5 seeds per system and cell.

V. Results

Figure 1 shows accuracy against h for each feature strength; Table I gives the $\mu = 1$ cells with seed standard deviations; Tables II and III give the winner and the GCN-MLP gap in all 33 cells. The MLP row is constant in h because, per seed, only the edges change.

H1: GCN below the MLP at mid homophily (partly supported, not where we registered it). At $\mu = 1$, GCN is below the MLP at $h = 0.3$ (0.449 vs 0.539, $p = 0.007293$) and at $h = 0.4$ the mean is also lower (0.506) but the paired test is not significant ($p = 0.1424$); at $h = 0.5$ GCN is above the MLP (0.638, $p = 0.01355$). By the registered criterion (negative and $p < 0.05$ in the cells 0.3, 0.4, 0.5) only one of three cells qualifies, so H1 is supported at $h = 0.3$ and not as a statement about the whole band. The deficit is in fact larger below the registered cells: Table III gives -0.09519 at $h = 0.2$ ($p = 0.0007776$ in Table I) and -0.09074 at $h = 0.3$. It grows with feature strength, from -0.03222 at $\mu = 0.5$, $h = 0.2$ to -0.2270 and -0.2533 at $\mu = 2$ ($h = 0.2, 0.3$), consistent with the picture that mixing in neighbours of other classes destroys good features; we did not run an experiment isolating that mechanism. For $\mu = 0.5$ the dip is small (-0.03222 and -0.01852 at $h = 0.2$ and 0.3) and we regard it as weak evidence. GCN also beats the MLP at $h = 0$ (+0.1700 at $\mu = 1$): with three classes, every neighbour of a node is from a different class, so aggregated neighbourhoods remain class-dependent. This U-shape is the shape suggested by [4], but our model does not test their neighbourhood-distribution explanation.

H2: ego/neighbour separation removes the dip (refuted as stated). H2GCN-style is the best system in all cells with $h \leq 0.2$ (Table II) and is far ahead under heterophily: at

$h = 0$, $\mu = 1$ it reaches 0.911 versus 0.709 for GCN and 0.539 for MLP. At $h = 0.3$ – 0.4 , $\mu = 1$ it recovers the MLP level (0.535, 0.536 vs 0.539) but does not exceed it. The registered criterion, within 0.02 of max(MLP, GCN) in every cell, fails in 6 of 33 cells (Table IV), the largest at $\mu = 0.5$, $h = 0.6$ where it trails GCN by 0.07778, and one with strong features ($\mu = 2$, $h = 0.3$) where it trails the MLP (0.804 vs 0.830). So the separation reduces but does not remove the mid-homophily dip, and at moderate-to-high homophily the extra neighbour weights cost accuracy when training data are scarce relative to parameters (we did not isolate this).

H3: label propagation (refuted). LP is the best system only in the high-homophily corner (Table II): $h \geq 0.6$ at $\mu = 0.5$, $h \geq 0.9$ at $\mu = 1$ and $h = 1$ at $\mu = 2$. The first part of H3 (best only with weak features and $h \geq 0.8$) is false because LP also wins at $\mu = 1, 2$ in the extreme cells, although there its margin over GCN/H2GCN is tiny (e.g. 1.000 vs 0.997 and 0.998 at $h = 1$, $\mu = 1$; at $h = 1$ LP’s ± 0.000 is not a bug: with $h = 1$ the graph splits into class-pure components that the 20% training labels cover). It is below the MLP at $h \leq 0.4$, $\mu = 1$ (0.389 at $h = 0.4$ vs 0.539).

H4: crossover moves with informativeness (supported on the homophilous branch). Reading the crossover as the lowest $h \geq 1/3$ at which GCN’s mean exceeds the MLP’s, it is $h = 0.4$ for $\mu = 0.5$, 0.5 for $\mu = 1$ and 0.6 for $\mu = 2$ (Table III): the stronger the features, the more homophily GCN needs. At $\mu = 0.5$ and $h = 0.4$ the gap is +0.009630, inside seed noise, so the $\mu = 0.5$ crossover is uncertain by about one grid step, and the ordering has only three levels of μ . This is a descriptive statement at 0.1 resolution, and as noted in Section IV it uses a post-hoc restriction of the registered wording.

Where the methods fail. GCN: $h \in [0.2, 0.4]$ for $\mu \geq 1$. H2GCN-style: moderate homophily with weak features (Table IV). LP: any $h \leq 0.5$ at $\mu \geq 1$, and it uses no features. MLP: wins only at $h = 0.3$ – 0.4 , $\mu \geq 1$ (Table II), the band where the graph is almost uninformative; its win over H2GCN-style at $\mu = 1$ and $h = 0.3$ (0.539 vs 0.535) is within noise.

VI. Ablations and Analysis

Ego/neighbour separation (H5). Table V compares the full H2GCN-style model (main-grid rows) with tied self/neighbour weights and with an added 2-hop term at $\mu = 1$. Tying the weights is catastrophic under heterophily: 0.724 drops to 0.356 at $h = 0.1$ ($p = 4.769 \times 10^{-6}$ in the paired test, 5 seeds) and also costs accuracy at $h = 0.9$ (0.994 to 0.930). At $h = 0.4$ it makes no measurable difference (0.536 vs 0.547, $p = 0.5502$). H5 is therefore supported at $h = 0.1$ and not supported at $h = 0.4$; with tied weights the model can only add or average neighbour features with the ego features, so it cannot flip the sign of neighbour information, which is consistent with the heterophilous failure but not a

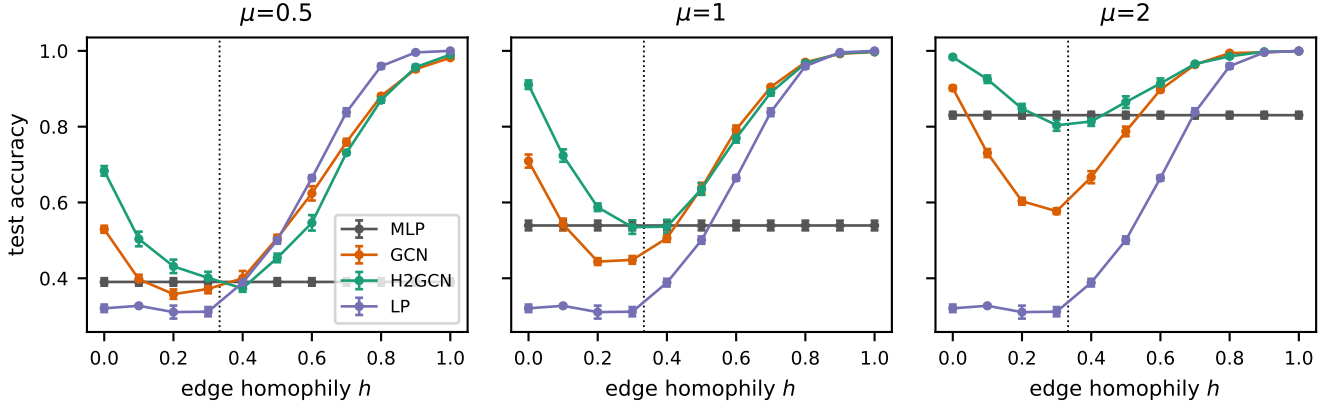


Fig. 1. Test accuracy versus edge homophily for the four systems at three feature strengths (mean over 5 seeds, bars show the standard error). The dotted line marks $h = 1/K$.

TABLE II
Best system (by mean over 5 seeds) in each of the 33 cells.

$\mu \setminus h$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.5	H2GCN	H2GCN	H2GCN	H2GCN	GCN	GCN	LP	LP	LP	LP	LP
1	H2GCN	H2GCN	H2GCN	MLP	MLP	GCN	GCN	GCN	GCN	LP	LP
2	H2GCN	H2GCN	H2GCN	MLP	MLP	H2GCN	H2GCN	H2GCN	GCN	H2GCN	LP

TABLE III
GCN minus MLP test accuracy (difference of the means over 5 seeds, as a fraction); negative values are cells where GCN is below the MLP.

$\mu \setminus h$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.5	0.1389	0.007037	-0.03222	-0.01852	0.009630	0.1156	0.2337	0.3689	0.4904	0.5615	0.5915
1	0.1700	0.002963	-0.09519	-0.09074	-0.03333	0.09889	0.2530	0.3652	0.4307	0.4530	0.4578
2	0.07148	-0.1000	-0.2270	-0.2533	-0.1637	-0.04296	0.06704	0.1333	0.1637	0.1659	0.1689

TABLE IV
Cells where H2GCN-style is more than 0.02 below the better of MLP and GCN (test accuracy, mean over 5 seeds).

μ	h	MLP	GCN	H2GCN
0.5	0.4	0.390	0.400	0.373
0.5	0.5	0.390	0.506	0.454
0.5	0.6	0.390	0.624	0.546
0.5	0.7	0.390	0.759	0.731
1	0.6	0.539	0.792	0.770
2	0.3	0.830	0.577	0.804

separately tested mechanism. Adding 2-hop neighbours helps slightly at $h = 0.1$ and $h = 0.9$ (see Δ in the table, nominal $p < 0.05$ each, not corrected for the six tests in this table) and does not at $h = 0.4$; with 5 seeds we call this suggestive only.

Mean-degree sensitivity. Table VI and Figure 2 vary the mean degree at the cell $h = 0.4$, $\mu = 1$. GCN stays below the MLP at all four degrees, with GCN-MLP equal to -0.05889 , -0.08444 , -0.03333 and -0.04815 (for $d = 2, 5, 10, 20$; the $d = 10$ cell is the non-significant main-grid

cell, the others have $p < 0.01$ uncorrected). H2GCN-style is level with the MLP at $d = 10$ and $d = 20$ in the means (0.536 vs 0.539 , 0.604 vs 0.602) and below it at $d = 2$ and $d = 5$ (0.527 vs 0.584 , 0.550 vs 0.583), so the picture that denser graphs rescue GCN at this homophily is not supported here; the degree trend is non-monotone and we do not interpret it.

VII. Limitations

This is a small CPU study of one synthetic family. (i) Scale: $n = 900$, 3 classes, 16-d Gaussian features, one mean degree for the main grid, two layers, hidden size 32; conclusions about where crossovers lie depend on these and need not transfer to larger or real graphs. (ii) Seeds: 5 seeds per cell, so cell-level p -values are noisy; the tests are uncorrected for the many cells examined and the verdicts we call supported or refuted at single cells (e.g. $h = 0.4$, $\mu = 1$) could flip with more seeds. (iii) Reimplemented baselines: our H2GCN-style model is a simplified ego/neighbour version without the intermediate-representation concatenation of the published method, our GCN and LP are reimplemented, and nothing was checked

TABLE V

Ablation of H2GCN-style at $\mu = 1$ (test accuracy mean $\pm$ std over 5 seeds; p from a paired t -test against the full model; Δ is the full model minus the variant, so a negative Δ means the variant is better).

h	H2GCN	tied	Δ	p	+2-hop	Δ	p
0.1	0.724 $\pm$ 0.037	0.356 $\pm$ 0.018	0.3681	4.769×10^{-6}	0.755 $\pm$ 0.020	-0.03148	0.03233
0.4	0.536 $\pm$ 0.040	0.547 $\pm$ 0.015	-0.01111	0.5502	0.524 $\pm$ 0.023	0.01259	0.3630
0.9	0.994 $\pm$ 0.003	0.930 $\pm$ 0.007	0.06333	3.984×10^{-5}	0.997 $\pm$ 0.002	-0.002963	0.03492

TABLE VI

Mean-degree sweep at $h = 0.4$, $\mu = 1$ (accuracy mean $\pm$ std, 5 seeds).

degree d	2	5	10	20
MLP	0.584 $\pm$ 0.009	0.583 $\pm$ 0.032	0.539 $\pm$ 0.029	0.602 $\pm$ 0.016
GCN	0.525 $\pm$ 0.023	0.499 $\pm$ 0.030	0.506 $\pm$ 0.023	0.554 $\pm$ 0.037
H2GCN	0.527 $\pm$ 0.020	0.550 $\pm$ 0.038	0.536 $\pm$ 0.040	0.604 $\pm$ 0.037
GCN-MLP	-0.05889	-0.08444	-0.03333	-0.04815
p (paired)	0.006717	0.003772	0.1424	0.008117

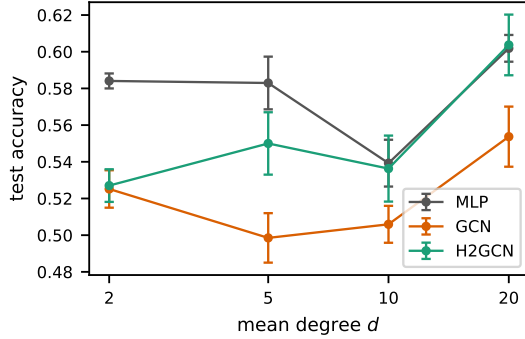


Fig. 2. Accuracy versus mean degree at $h = 0.4$, $\mu = 1$.

against the authors’ code. (iv) Tuning: a 8-configuration grid on separate graphs at three cells with one feature strength, shared across all h and μ ; per-cell tuning, other depths or hidden sizes could change rankings, especially the MLP/GCN margin at mid homophily. (v) Model: uniform inter-class edge probabilities make heterophily maximally informative about class (neighbours are always another class), which flatters models that can use neighbour information; structured inter-class edges, degree heterogeneity and class imbalance are not covered. (vi) Mechanisms: we did not run experiments isolating why GCN loses to the MLP or why tied weights fail, so those explanations are hypotheses. (vii) Post-hoc choice: the crossover reading used for H4 was clarified after seeing the data. A full study would add real heterophilous benchmarks [6], more seeds, other model families [3], [10], and per-cell tuning.

VIII. Conclusion

On contextual SBM graphs, whether message passing helps depends on both homophily and feature strength. GCN beats the MLP at high homophily and, in this three-

class model, also at $h = 0$; it falls below the MLP in a low-to-mid homophily band that, at $\mu = 1$, is significant at $h = 0.2$ and 0.3 but not at $h = 0.4$, and that deepens with stronger features. Separating self and neighbour weights fixes most of the heterophilous failure but not uniformly (6 of 33 cells fall more than 0.02 short of the better of MLP and GCN), and label propagation wins only in the high-homophily corner. These results come from 5 seeds on one small synthetic family with reimplemented baselines and should be treated as a controlled map, not a general rule; extending to more seeds, larger graphs and real benchmarks is the obvious next step.

References

- [1] T. N. Kipf and M. Welling, “Semi-Supervised Classification with Graph Convolutional Networks,” arXiv preprint arXiv:1609.02907, 2016. [Online]. Available: <https://arxiv.org/abs/1609.02907>
- [2] J. Zhu, Y. Yan, L. Zhao, M. Heimann, L. Akoglu, and D. Koutra, “Beyond Homophily in Graph Neural Networks: Current Limitations and Effective Designs,” arXiv preprint arXiv:2006.11468, 2020. [Online]. Available: <https://arxiv.org/abs/2006.11468>
- [3] E. Chien, J. Peng, P. Li, and O. Milenkovic, “Adaptive Universal Generalized PageRank Graph Neural Network,” arXiv preprint arXiv:2006.07988, 2020. [Online]. Available: <https://arxiv.org/abs/2006.07988>
- [4] Y. Ma, X. Liu, N. Shah, and J. Tang, “Is Homophily a Necessity for Graph Neural Networks?” arXiv preprint arXiv:2106.06134, 2021. [Online]. Available: <https://arxiv.org/abs/2106.06134>
- [5] S. Luan, C. Hua, Q. Lu, J. Zhu, M. Zhao, S. Zhang, X.-W. Chang, and D. Precup, “Revisiting Heterophily For Graph Neural Networks,” arXiv preprint arXiv:2210.07606, 2022. [Online]. Available: <https://arxiv.org/abs/2210.07606>
- [6] O. Platonov, D. Kuznedelev, M. Diskin, A. Babenko, and L. Prokhorenkova, “A critical look at the evaluation of GNNs under heterophily: Are we really making progress?” arXiv preprint arXiv:2302.11640, 2023. [Online]. Available: <https://arxiv.org/abs/2302.11640>
- [7] Y. Deshpande, A. Montanari, E. Mossel, and S. Sen, “Contextual Stochastic Block Models,” arXiv preprint arXiv:1807.09596, 2018. [Online]. Available: <https://arxiv.org/abs/1807.09596>
- [8] A. Baranwal, K. Fountoulakis, and A. Jagannath, “Graph Convolution for Semi-Supervised Classification: Improved Linear Separability and Out-of-Distribution Generalization,” arXiv preprint arXiv:2102.06966, 2021. [Online]. Available: <https://arxiv.org/abs/2102.06966>
- [9] K. Oono and T. Suzuki, “Graph Neural Networks Exponentially Lose Expressive Power for Node Classification,” arXiv preprint arXiv:1905.10947, 2019. [Online]. Available: <https://arxiv.org/abs/1905.10947>
- [10] D. Lim, F. Hohne, X. Li, S. L. Huang, V. Gupta, O. Bhalerao, and S.-N. Lim, “Large Scale Learning on Non-Homophilous Graphs: New Benchmarks and Strong Simple Methods,” arXiv preprint arXiv:2110.14446, 2021. [Online]. Available: <https://arxiv.org/abs/2110.14446>

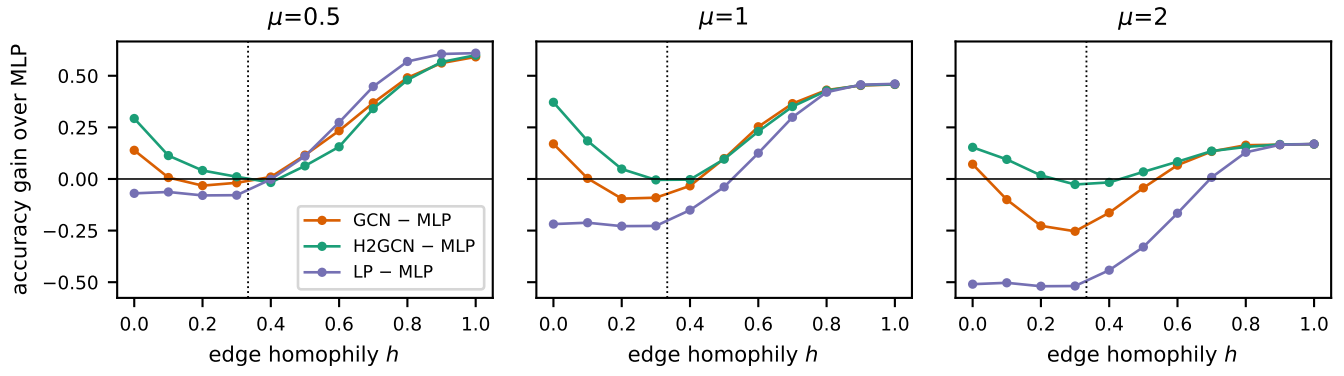


Fig. 3. Accuracy gain over the MLP for GCN, H2GCN-style and LP versus h (mean over 5 seeds). Values below zero mark where message passing hurts.

- [11] H. Pei, B. Wei, K. C.-C. Chang, Y. Lei, and B. Yang, "Geom-GCN: Geometric Graph Convolutional Networks," arXiv preprint arXiv:2002.05287, 2020. [Online]. Available: <https://arxiv.org/abs/2002.05287>